Response Letter: Diffusion-Driven Modeling of Pareto Set Evolution for Dynamic Multiobjective Optimization [SMCA-26-06-3271]

The authors sincerely appreciate the opportunity to revise the paper according to the comments made by the editor, senior editor, and reviewers. We would like to thank them for their highly valuable suggestions and comments. Responses to all suggestions and comments have significantly improved the quality and presentation of the manuscript. In particular, we have highlighted major modifications in red in the revision.

- Annotations:
  - **RC**: ***Reviewers' Comment***, AR: Authors' Response, ♢: *Revised manuscript Text*.
  - TABLE X and Fig. Y refer to TABLE X and Fig. Y in the manuscript.
  - TABLE S-X and Fig. S-Y refer to TABLE S-X and Fig. S-Y in the supplementary material.
  - Because some references are cited in the response letter, the citation order for the same reference may differ from that in the manuscript.

Senior Editor 1

RC: ***1. Old methods such as [48, 51] can be replaced by recent diffusion- or GAN-based methods. 2. Authors are asked to copy the results from the past publications. It is possible, if authors used the same problems with the same max_FEs. This will eliminate the need to re-program and re-tune. 3. In the captions of the tables of results, authors must clearly state whether results of competing methods were copied from the original publications. 4. If authors did not copy results, then authors must re-tune all methods in the same way authors tuned the proposed method (without using the default parameters of the compared methods).***

AR: Thank you very much for your valuable comments. In response to these comments, we made the following revisions and provided additional clarifications.

(1) We retained KT-DMOEA [1] and IGP-DMOEA [2] and further clarified the reasons for retaining them. Specifically, KT-DMOEA is a representative knee-point-driven dynamic multiobjective optimization method and is directly relevant to the idea of using knee points to predict target regions in this paper. IGP-DMOEA employs inverse Gaussian processes to model and predict dynamic populations and is representative of probability-model-driven dynamic response methods. Therefore, they are meaningful comparison algorithms for DD-DMOEA. Meanwhile, the current experiments have already included several recent methods, such as DIP-DMOEA, PSGN, VARE, and the diffusion-based DM-DMOEA. Accordingly, we have further explained the reasons for selecting these comparison algorithms in Section IV-B and the supplementary material.

(2) We did not directly copy the results from the original publications. For methods with publicly available source code, we obtained the results by running their public implementations. For methods without publicly available source code, we reimplemented them according to the original publications and conducted the experiments. We have also added the corresponding clarification in Section IV-B.

(3) We have added notes below Table I in the main manuscript and Table S-II in the supplementary material. Specifically, "†" indicates results obtained by running publicly available source code, while "‡" indicates results obtained from our reimplementations based on the original publications, thereby clearly specifying the source of the results for each comparison algorithm.

(4) We did not directly adopt their default parameter settings. Instead, the key parameters were re-tuned following the same parameter-tuning principle as DD-DMOEA to ensure the fairness of the comparison. The corresponding clarification has been added to Section IV-B.

♢: On page 8, we have made the following changes. *The proposed DD-DMOEA is compared with seven representative DMOEAs, including KT-DMOEA [1], IGP-DMOEA [2], KTM-DMOEA [3], DIP-DMOEA [4], PSGN [5], VARE [6], and DM-DMOEA [7]. Specifically, KT-DMOEA is a representative knee-point-driven method, while IGP-DMOEA employs inverse Gaussian process modeling for dynamic population prediction. The comparison also includes recent learning- and diffusion-based DMOEAs to provide a comprehensive evaluation. Details of these algorithms are provided in the supplementary material.*
Moreover, all comparison results were obtained by independently running publicly available source codes or our re-implementations based on the original papers. For a fair comparison, the key parameters of all re-run algorithms were re-tuned following the same principle as DD-DMOEA.

♢: On page 10 and on page 5 of the supplementary materials, we have made the following changes. ***Note:*** *Results marked with "†" were obtained by running the publicly available source code; results marked with "‡" were obtained using our reimplementation based on the original publications.*

Editor 2

RC: ***This is a revised manuscript. The reviewers feel that the manuscript is appropriate for the journal and most of the concerns are resolved. However, one of the reviewer further raised some major concerns that need to be addressed. The authors are asked to prepare a major revision.***

AR: In response to the issues raised by the reviewer in this round of review, we have made the following revisions.
(1) We further improved Algorithm 1 and Algorithm 2 by clarifying the generation of the relevant populations and the inputs and outputs of the algorithms, and unified the symbol definitions in Eq. (7), Eq. (8), and the pseudocode.
(2) We further clarified the meaning, applicable conditions, and limitations of "noisy POS", and clarified the distinction between "diffusion-driven" and conventional generative diffusion models.
(3) We supplemented the complete parameter sensitivity settings for N_s and further clarified the sources of the results for the comparison algorithms.
(4) We systematically checked and corrected the terminology, tables, and related language expressions throughout the manuscript and supplementary material.

Reviewer #1

RC: ***What are the additional ways in which the paper could be improved: This is a revised paper. The authors have addressed all of my comments successfully. This paper could be accepted in the current form.***

AR: Thank you very much for your positive evaluation of our revised manuscript. We sincerely appreciate your valuable comments and suggestions, which have helped us further improve the quality of the manuscript.

Reviewer #2

RC: ***What are the additional ways in which the paper could be improved: The authors have addressed all major concerns raised during review.***

AR: Thank you very much for your positive evaluation of our revised manuscript. We sincerely appreciate your valuable comments and suggestions, which have helped us further improve the quality of the manuscript.

Reviewer #3

Comment #3.1

RC: ***"The crossover operator and the variance operator in MOEA/D. . . . . " should be "The crossover operator and the variation operator in MOEA/D. . . . . ".***

AR: Thank you very much for your valuable comments. We have checked and corrected this terminology by replacing the inaccurate term "variance operator" with "mutation operator". We have also further checked the relevant expressions throughout the manuscript to ensure the accuracy and consistency of the terminology.

♢: On page 8, we have made the following changes. *The crossover operator and the mutation operator in MOEA/D are kept the same in all the compared algorithms.*

Comment #3.2

RC: ***Table S-I still contains several language and formatting errors. For example, "mixed convexity-concavity, location of optima" should be revised to"mixed convexity-concavity, location of optima" by adding a space after the comma. In addition, the phrase "local of optima" in the DF5 row is grammatically incorrect and should be corrected to "location of optima".***

AR: Thank you very much for your valuable comments. We have checked and corrected the formatting issues in Table S-I and replaced the inaccurate phrase "local of optima" in DF5 with "location of optima". Meanwhile, we have also uniformly checked and revised the other descriptions in this table.

♢: On page 3 of the supplementary materials, we have made the following changes:

TABLE I
DYNAMIC CHARACTERISTICS OF 14 TEST FUNCTIONS IN THE CEC2018 DYNAMIC MULTIOBJECTIVE OPTIMIZATION COMPETITION.

Problem	Objectives	Dynamics	Remarks
DF1	2	mixed convexity-concavity, location of optima	dynamic PF and PS
DF2	2	switch of position-related variable, location of optima	static convex PF, dynamic PS, severe diversity loss
DF3	2	mixed convexity-concavity, variable-linkage, location of optima	dynamic PF and PS
DF4	2	variable-linkage, PF range, bounds of PS	dynamic PF and PS
DF5	2	number of knee regions, location of optima	dynamic PF and PS
DF6	2	mixed convexity-concavity, multimodality, location of optima	dynamic PF and PS
DF7	2	PF range, location of optima	convex PF, static PS centroid, dynamic PF and PS
DF8	2	mixed convexity-concavity, distribution of solutions, location of optima	static PS centroid, dynamic PF and PS, variable-linkage
DF9	2	number of disconnected PF segments, location of optima	dynamic PS and PF, variable-linkage
DF10	3	mixed convexity-concavity, location of optima	dynamic PS and PF, variable-linkage
DF11	3	size of PF region, PF range, location of optima	dynamic PS and PF, concave PF, variable-linkage
DF12	3	number of PF holes, location of optima	dynamic PS, static concave PF, variable-linkage
DF13	3	number of disconnected PF segments, location of optima	dynamic PS and PF, the PF can be a continuous convex or concave segment, or several disconnected segments
DF14	3	degenerate PF, number of knee regions, location of optima	dynamic PS and PF, variable-linkage

Comment #3.3

RC: ***Although the authors provide an explanation in the response letter and the surrounding text, Algorithm 1 itself still does not explicitly define how these two populations are generated. In Line 9, the algorithm directly constructs*** $initPop$***, but neither*** $initPop$ ***nor has been initialized or assigned in the pseudocode. For a rigorous algorithmic description, the authors should add explicit steps. In addition, the manuscript should specify the population sizes of*** Pop_{pred}***,*** Pop_{last}***, and*** Pop_{rand}***, and clarify how the final*** $initPop$ ***is adjusted to satisfy the prescribed population size. Without these details, Algorithm 1 is not fully reproducible.***

AR: Thank you very much for your valuable comments. We have added explicit steps in Algorithm 1 to define the historical population Pop_{last} and the random population Pop_{rand}, respectively, and clarified the sizes of the three populations and the construction of the final $initPop$ in Section III-A.

Specifically, both Pop_{pred} and Pop_{rand} have a population size of N, while Pop_{last} is composed of POS_{t-1} obtained from the previous environment. After merging the three populations, N individuals are selected through nondominated sorting to form the final $initPop$.

♢: On page 4, we have made the following changes. *The pseudocode of the proposed DD-DMOEA is presented in Algorithm 1. Initially, the relevant parameters and the discrete-time environment variable $t = 1$ are set. Subsequently, the algorithm enters the main loop at line 2, which corresponds to the dynamic response phase. In this process, if the discrete-time variable $t \leq 2$, the algorithm randomly initializes a population $initPop$ for the current environment in line 4, and obtains an approximate POS (POS_t) at environment moment t by the static MOEA (SMOEA) in line 5. Note that the SMOEA used in this paper is MOEA/D. Instead, the algorithm executes lines 7-12, which are also the core of the algorithm in this paper. First, the AKP strategy in line 7 predicts the knee point information $knees$ at the current moment t based on the knee points ($knee_{t-1}$ and $knee_{t-2}$) of the previous two moments. Then, in line 8, the DDM considers the POS_{t-1} at moment t-1 to be a noisy distribution, through an efficient and progressive denoising process, and generates high-quality solutions (Pop_{pred}) adapted to the environment at moment t. To maintain diversity, three populations are constructed: the DDM-predicted population (Pop_{pred}) of size N, the historical population (Pop_{last}) containing the POS_{t-1}, and the random population (Pop_{rand}) of size N. The size of Pop_{last} depends on the obtained POS, and the three populations are merged and subjected to nondominated sorting to select N individuals for $initPop$. Finally, the approximate POS at the current moment (POS_t) is obtained by SMOEA in line 12. Since the knee point information of AKP is obtained by prediction, the true knee point information at moment t is evaluated to satisfy the iterative requirement in line 14. Given that the guidance of DDM is highly dependent on the accuracy of the predicted knee points, this paper adopts an uncertainty-aware diffusion guidance strategy at line 16, thereby enhancing the robustness of the algorithm.*

Algorithm 1 DD-DMOEA

Input: The dynamic problem $F(\boldsymbol{x}, t)$, population size N, the number of subspaces N_s.
Output: An approximate POS and POF in different environments.

1: Initialize related parameters and set t=1;
2: **while** *the environment has changed* **do**
3: **if** $t \leq 2$ **then**
4: Randomly initialize a population *initPop*;
5: POS_t=**SMOEA**(*initPop*, $F(\boldsymbol{x}, t)$);
6: **else**
7: $knees$=**AKP**($knee_{t-1}$, $knee_{t-2}$, $F(\boldsymbol{x}, t)$);
8: Pop_{pred}=**DDM**(POS_{t-1}, $knees$, N_s, ψ^t);
9: $Pop_{last} \leftarrow$ historical POS_{t-1};
10: $Pop_{rand} \leftarrow$ a randomly generated population;
11: *initPop*=$Pop_{pred} \cup Pop_{last} \cup Pop_{rand}$;
12: POS_t=**SMOEA**(*initPop*, $F(\boldsymbol{x}, t)$);
13: **end if**
14: Evaluate the true knee points at the moment t by POS_t;
15: $t = t + 1$;
16: Calculate the adaptive variance ψ^t of the current Gaussian distribution;
17: **end while**

Comment #3.4

RC: ***The description of Algorithm 2 still contains several inconsistencies that affect reproducibility. In particular, Line 1 randomly initializes a population, but this population is never used in the subsequent steps. The algorithm directly obtains ($knee_{t-1}$) and ($knee_{t-2}$) from (POS_{t-1}) and***

(POS_{t-2}), and then predicts the current knee points through direction-vector and deflection-angle calculations. Therefore, the role of the randomly initialized population is unclear. If this population is intended to assist the estimation of knee points in the current environment, the authors should explicitly describe how it is evaluated and how it contributes to the AKP procedure. Otherwise, Line 1 should be removed, and the input parameter (N) should also be deleted from Algorithm 2 because it is not used. In addition, the output of Algorithm 2 should be stated more precisely. The current output, "Knee points in the new environments", is vague. Since the algorithm returns the set of knees, the output should be written as the predicted knees for the current environment.

AR: Thank you very much for your valuable comments. We have carefully checked and revised Algorithm 2. In addition, we have further clarified the inputs and outputs of Algorithm 2, such that the historical knee points are directly obtained based on POS_{t-1} and POS_{t-2}, and the output is explicitly defined as the predicted knee points for the current environment. This makes the algorithmic description consistent with the AKP procedure in the main text.

♢: On page 5, we have made the following changes.

Algorithm 2 Auxiliary Strategy Based on Knee Point Estimation (AKP)

Input: The dynamic problem $F(\boldsymbol{x}, t)$, the POS_{t-1} and POS_{t-2} at moments t-1 and t-2, the number of subspaces N_s.

Output: Predicted knee points in the new environments.

Obtain the $knee_{t-1}$, $knee_{t-2}$ by POS_{t-1}, POS_{t-2} respectively;
for i=1 *to* N_s **do**
  for j=1 *to* d-1 **do**
    Obtain the direction vectors v_i;
    Calculate the j-th angular coordinates β_i^j of v_i through Eq.(5);
    Calculate the deflection angles θ_i^j via Eq.(6);
  end for
  The knee points $knee_t^i$ and Θ_i are obtained through Eq.(7) and (8), respectively;
end for
$knees = \{knee^i \mid i \in [1, N_s]\}$;
return $knees$;

Comment #3.5

RC: ***The current form of Eq. (7) is ambiguous. A more consistent form would be $knee_t^i = knee_{t-1}^i + r_i \Theta_i$. In addition, Eq. (8) should be made consistent with the dimensional notation used in the preceding text and Algorithm 2. The angular coordinates are defined for ($j \in [1, d-1]$), and Algorithm 2 also iterates from (j=1) to (d-1). However, Eq. (8) uses n in the conditions, while the text states that (d) is the size of the decision vector.***

AR: Thank you very much for your valuable comments. We have checked and unified the notation in Eq. (7), Eq. (8), and Algorithm 2. Meanwhile, the dimensional notation in Eq. (8) has been uniformly changed to d, which is consistent with the definition of d as the dimension of the decision vector in the preceding text and with $j = 1, \ldots, d-1$ in Algorithm 2.

♢: On page 6, we have made the following changes.

$$knee_t^i = knee_{t-1}^i + r_i \times \Theta_i. \tag{1}$$

$$\Theta_i^j = \begin{cases} \cos(\beta_i^j + \theta_i^j), & j = 1 \\ \prod_{p=1}^{j-2} \sin(\beta_i^p + \theta_i^p) \cos(\beta_i^{j-1} + \theta_i^{j-1}), & 1 < j < d-1 \\ \prod_{p=1}^{j-1} \sin(\beta_i^p + \theta_i^p), & j = d-1 \end{cases}. \tag{2}$$

Comment #3.6

RC: ***In page 11 of 63, the authors' statement that the sensitivity test only uses is incomplete. The tested parameter set should be reported as shown in Fig. S-12.***

AR: Thank you very much for your valuable comments. We have supplemented the complete parameter settings for the N_s sensitivity experiment. Specifically, under the dynamic environment with $n_t = 5$ and $\tau_t = 5$, N_s was set to $\{3, 5, 7, 10, 15\}$ for comparison, and the MIGD and MHV results under different settings were analyzed using the spider plots in Fig. S-12.

$\diamondsuit$: On page 11, we have made the following changes. *To investigate the sensitivity to the number of subspaces, N_s was set to $\{3, 5, 7, 10, 15\}$ under $n_t = 5$ and $\tau_t = 5$. Fig. S-12 presents the MIGD and MHV results as spider plots, where the outermost scale represents the smallest MIGD and the largest MHV. The results show that $N_s = 5$ achieves the best overall performance. A smaller N_s limits accurate target-region partitioning and knee-point identification, whereas a larger N_s causes excessive fragmentation, destabilizes knee-point prediction, and increases computational costs.*

Comment #3.7

RC: ***The authors should clarify whether the Gaussian guidance density is evaluated in the objective space or the decision space. AKP predicts knee points through objective-space partitioning, while DDM applies to historical solutions. If $knee^i$ is an objective-space knee point and v_i is a decision-space vector, the density is dimensionally inconsistent. If $knee^i$ is the corresponding decision-space knee solution, the mapping from the objective-space knee region to the decision-space representative must be explicitly described.***

AR: Thank you very much for your valuable comments. In the proposed AKP strategy, only the subspace partitioning is performed in the objective space. Specifically, the objective value is used to assign each POS to an objective-space subregion. AKP then selects a knee point within each subregion, and the decision vector of that solution is denoted by $knee_t^i$. Thus, $knee_t^i$, its predicted value $\mu_i = knee_t^i$, and every historical sample used by DDM are all d-dimensional decision vectors.

Consequently, the Gaussian guidance density, $f_i(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x}; \mu_i, \psi_i^2)$, is evaluated entirely in the decision space and is dimensionally consistent. No inverse mapping from a predicted objective-space knee point to a decision-space vector is required, because each objective vector remains associated with the decision vector that generates it. Objective-space partitioning is used only to group solutions and maintain subregion correspondence across consecutive environments.

We have revised Sections III-A and Algorithm 3 to make this distinction explicit. We have also unified the notation for the decision-space dimension and clarified that the knee-prediction error in Eq. (16) is calculated in the decision space.

$\diamondsuit$: On page 4, we have made the following changes. *we introduce an auxiliary strategy based on knee-point estimation (AKP). AKP first partitions the objective space according to the objective vectors of the historical solutions. Within each subregion, it identifies a representative solution and defines its decision vector as the knee point. The subsequent population prediction is performed in the decision space. Therefore, objective-space partitioning is used only to organize the historical solutions and establish regional correspondence. Subsequently, in the decision space, a diffusion-driven dynamic response mechanism (DDM) utilizes these representative points as reference information.*

$\diamondsuit$: On page 5, we have made the following changes. *Subsequently, the decision-space knee vectors obtained from the two preceding environments, $knee_{t-1}$ and $knee_{t-2}$, are used to characterize the movement of the i-th knee solution. The corresponding direction vector is $v_i = knee_{t-1}^i - knee_{t-2}^i$. Ultimately, a trend prediction model (TPM) [1] is constructed by tracking these motion trajectories across consecutive environments. TPM is therefore constructed entirely in the decision space.*

$\diamondsuit$: On page 6, we have made the following changes. *The dynamic problem $F(\boldsymbol{x}, t)$, the POS_{t-1} at moment t-1, number of subspaces N_s, size of sample set S in the i-th subspace n_i, total time step K, knee point predicted by the AKP in the i-th subspace $\mu_i = knee^i$, adaptive standard deviation ψ_i.*

$\Diamond$: On page 7, we have made the following changes. *Here, $f_i(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x}; \mu_i, \psi_i^2)$, where $\mu_i = knee^i$ is the decision-space knee point predicted by the AKP, and ψ is the adaptive standard deviation. Details are provided in Subsection III-D. $f_i(\boldsymbol{x})$ represents the Gaussian probability density guided by the predicted knee point in the i-th subspace.*
We first define a decision-space prediction error as the Euclidean distance between the predicted knee point $knee_{pred}^{i,t-1}$ at time $t-1$ and the true knee point $knee_{true}^{i,t-1}$ obtained after optimization.

Comment #3.8

RC: ***POS_{t-1} is a high-quality optimal solution set in the previous environment, although it may become misaligned with the current environment. The authors should clarify under what dynamic conditions POS_{t-1} can be reasonably regarded as a noisy version of POS_t. It is essential to explicitly state the assumptions under which the "noisy POS" interpretation is valid and to discuss how the proposed DDM behaves when these assumptions are violated.***

AR: Thank you very much for your valuable comments. We clarify that "noisy" describes the mismatch between the historical POS_{t-1} and the new environment. This interpretation does not require small environmental changes or highly similar consecutive POS. Instead, it assumes that POS_{t-1} retains sufficient transferable structural information to guide the population toward promising regions of the new POS_t.
We have also added a discussion of the limitations of this assumption. When environmental changes substantially reconstruct the feasible manifold or introduce highly discontinuous structures, the historical population may provide insufficient information for the new POS, thereby reducing the reliability of diffusion guidance. For example, DF12 represents such a case, where the time-varying holes in the latent PF weaken the correspondence between historical solutions and the feasible regions in the new environment. Accordingly, we have further clarified this assumption and its limitations in Section III-C, the experimental analysis of DF12, and the Conclusion.

$\Diamond$: On page 6, we have made the following changes.*Specifically, relative to the target POS_t, the historical POS_{t-1} is treated as a noisy sample set that retains useful structural information from the previous environment. Here, "noisy" describes the mismatch between POS_{t-1} and the new environment rather than random noise. This interpretation assumes that POS_{t-1} still retains sufficient transferable information to guide the population toward promising regions of the new POS_t, rather than requiring highly similar consecutive POS.*

$\Diamond$: On page 9, we have made the following changes. *For DF12, the number of holes in its latent PF changes over time, substantially altering the feasible manifold. In this case, the historical POS_{t-1} may provide insufficient information for some newly formed feasible regions, weakening the noisy-POS assumption. Consequently, the predicted target regions cannot accurately characterize the discontinuous manifold, and diffusion-based relocation may drive solutions into the hole regions.*

$\Diamond$: On page 12, we have made the following changes. *DD-DMOEA still has some limitations. First, DD-DMOEA relies on continuous probability density to guide population migration, and therefore still has limitations when dealing with problems with discontinuous PF structures. When environmental changes substantially reconstruct the feasible manifold or introduce highly discontinuous PF/POS structures, this assumption may be weakened, thereby reducing the reliability of diffusion guidance. Future work will explore structure-aware modeling to better handle discontinuous fronts. Second, DD-DMOEA relies on the accuracy of knee point prediction. Under drastic, nonlinear, or irregular changes, prediction errors may weaken the guidance effect. Future work can utilize multi-source historical data to explore more robust target estimation methods. Finally, the proposed framework will be extended to more complex scenarios, including higher-dimensional objective spaces, constrained DMOPs, and real-world applications, to further evaluate its scalability and practical potential.*

Comment #3.9

RC: ***Although the authors clarify that DDM is different from standard generative diffusion models, the manuscript still frequently uses strong terms such as "diffusion mechanism", "diffusion-***

driven", and "denoising diffusion process". These terms may overstate the theoretical connection to diffusion models, because DDM does not define a genuine forward noising process or learn a reverse score/noise predictor. The method appears more accurately characterized as a diffusion-inspired, knee-point-guided multi-step population relocation strategy. The authors should use this terminology consistently to avoid misleading readers.

AR: Thank you very much for your valuable comments. We have further revised several expressions, particularly "denoising diffusion process" and "reverse denoising", to avoid implying that DDM is a conventional generative diffusion model.

However, we retain the term "diffusion-driven" because the dynamic response process of DDM is designed based on the progressive state-transformation principle of diffusion models. Specifically, POS_{t-1} is regarded as a perturbed historical state relative to the new environment, and DDM progressively transforms this state toward the target regions through multiple intermediate states. Its multi-step update structure is therefore driven by the underlying idea of diffusion-based progressive transformation, although the target estimate and perturbation terms are obtained analytically rather than learned by a neural network. In this sense, "diffusion-driven" describes the design principle of the dynamic response mechanism rather than claiming that DDM itself is a standard generative diffusion model.

To avoid potential misunderstanding, we have explicitly added this definition when "diffusion-driven" is introduced and further clarified the distinction between DDM and conventional diffusion models in Section III-C. We have also revised potentially misleading expressions such as "denoising diffusion process" throughout the Introduction, method description, Fig. 2, and Conclusion, while retaining "diffusion-driven" as the terminology describing the underlying design principle of DD-DMOEA.

$\Diamond$: On page 2, we have made the following changes. *By conceptualizing the solution evolution as a diffusion-driven multi-step transformation process, we avoid the complexity of traditional deep learning models, offering a more efficient and scalable solution.*

$\Diamond$: On page 5, we have made the following changes. ***(b)** Guided by the predicted knee points $knee_t$ and historical POS_{t-1}, the DDM performs a K-step diffusion-driven analytical transformation process to progressively generate the predicted POS_t for the new environment.*

$\Diamond$: On page 6, we have made the following changes. *Therefore, "diffusion-driven" in this paper refers to a dynamic response process driven by the diffusion-inspired principle of progressive state transformation.*

Inspired by the progressive reverse denoising of diffusion models, DDM adopts a multi-step analytical update structure to characterize population relocation across intermediate states.

$\Diamond$: On page 7, we have made the following changes. *The core of this strategy is to reframe the evolution from POS_{t-1} to POS_t as a diffusion-driven multi-step transformation from a perturbed historical state toward the target state.*

$\Diamond$: On page 12, we have made the following changes. *The proposed method regarded the POS in consecutive environments as different states in a diffusion process and modeled the dynamic response as a diffusion-driven analytical multi-step population transformation process, thereby enabling a more natural characterization of the gradual evolutionary trend of the POS while avoiding the response delay caused by model training.*

REFERENCES

[1] M. Jiang, Z. Wang, H. Hong, and G. G. Yen, "Knee point-based imbalanced transfer learning for dynamic multiobjective optimization," *IEEE Transactions on Evolutionary Computation*, vol. 25, no. 1, pp. 117–129, 2021.

[2] H. Zhang, J. Ding, M. Jiang, K. C. Tan, and T. Chai, "Inverse gaussian process modeling for evolutionary dynamic multiobjective optimization," *IEEE Transactions on Cybernetics*, vol. 52, no. 10, pp. 11 240–11 253, 2022.

[3] Q. Lin, Y. Ye, L. Ma, M. Jiang, and K. C. Tan, "Dynamic multiobjective evolutionary optimization via knowledge transfer and maintenance," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 54, no. 2, pp. 936–949, 2024.

[4] Y. Ye, S. Liu, J. Zhou, Q. Lin, M. Jiang, and K. C. Tan, "Learning-based directional improvement prediction for dynamic multiobjective optimization," *IEEE Transactions on Evolutionary Computation*, pp. 1–1, 2024.

[5] W. Song, S. Liu, X. Wang, Y. Guo, S. Yang, and Y. Jin, "Learning to guide particle search for dynamic multiobjective optimization," *IEEE Transactions on Cybernetics*, vol. 54, no. 9, pp. 5529–5542, 2024.
[6] S. Jiang, Y. Wang, Y. Hu, Q. Zhang, and S. Yang, "Dynamic multi-objective optimisation based on vector autoregressive evolution," *IEEE Transactions on Evolutionary Computation*, pp. 1–1, 2025.
[7] F. Wang, J. Xie, A. Zhou, and K. Tang, "A new prediction strategy for dynamic multi-objective optimization using diffusion model," *IEEE Transactions on Evolutionary Computation*, pp. 1–1, 2025.